Lecture Ten

1 Applications of Differentiation

1.1 Concavity, Points of Inflection and The Second Derivative Test

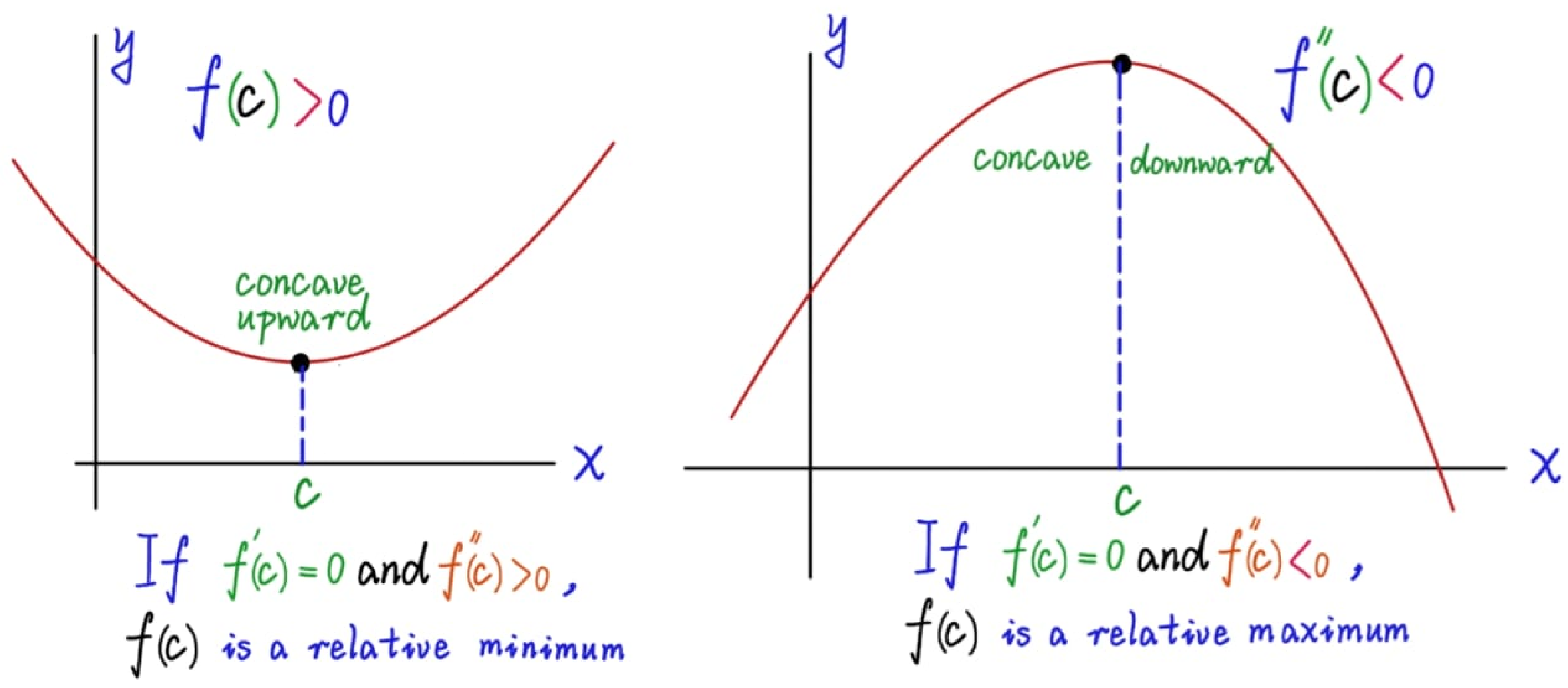
In this section we will see how locating the interval in which f' increases or decreases can be used to determine where the graph of f is curving upward or curving downward.

Definition 1.1.1 *The graph of a differentiable function $y = f(x)$ is*

(a) ***Concave up*** *on an open interval I if f' is increasing on I.*

(b) ***Concave down*** *on an open interval I if f' is decreasing on I.*

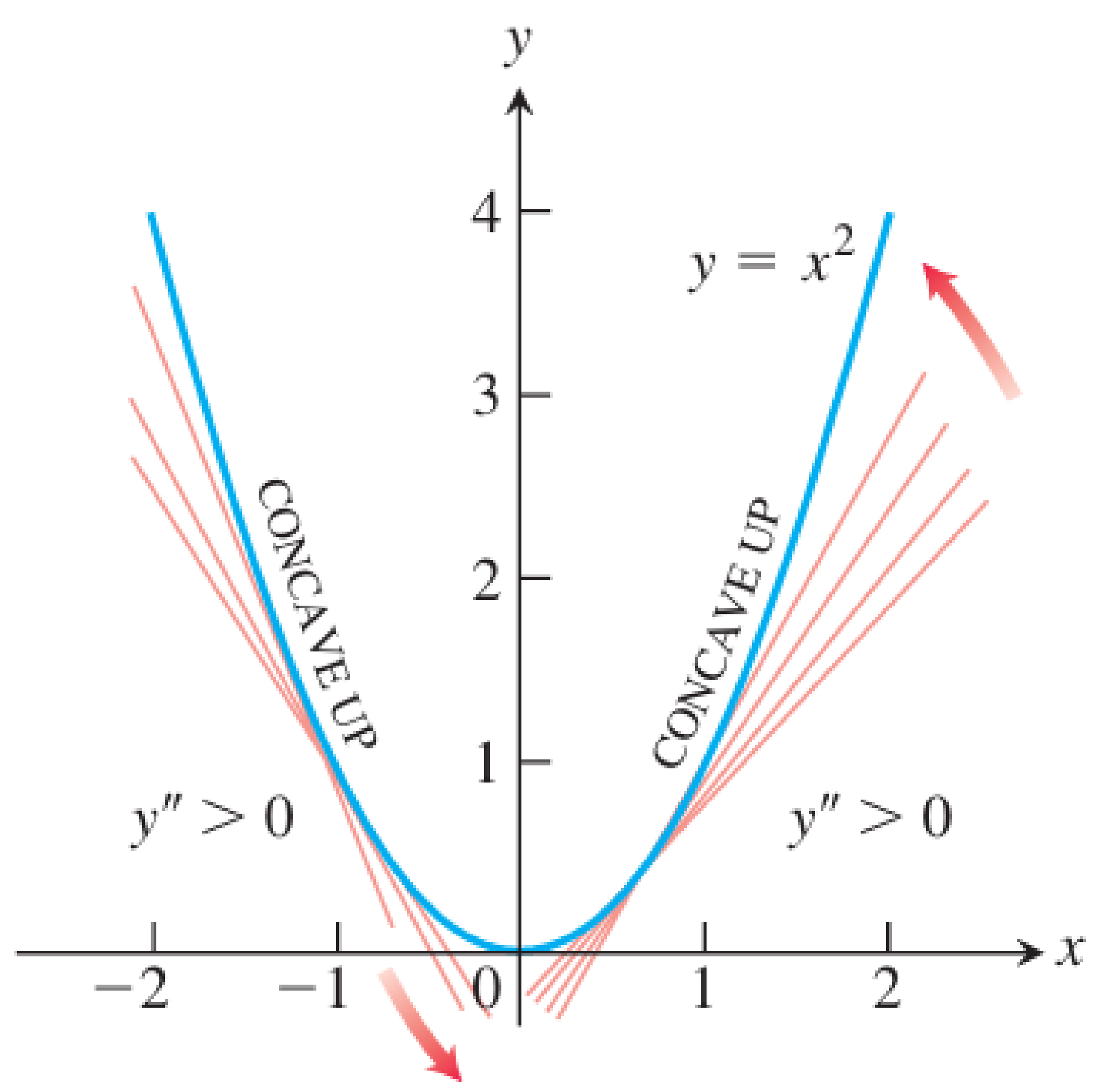
Note: *A function whose graph is concave up is also often called* ***Convex****.*



To find then open intervals on which the graph of a function f is concave upward or downward you need to find the intervals on which f' is increasing or decreasing.

Example 1.1.1 *1. The curve $y = x^3$ is concave down on $(-\infty, 0)$ where $y'' = 6x < 0$ and concave up on $(0, \infty)$ where $y'' = 6x > 0$.*

2. The curve $y = x^2$ is concave up on $(-\infty, \infty)$ because its second derivative $y'' = 2$ is always positive.

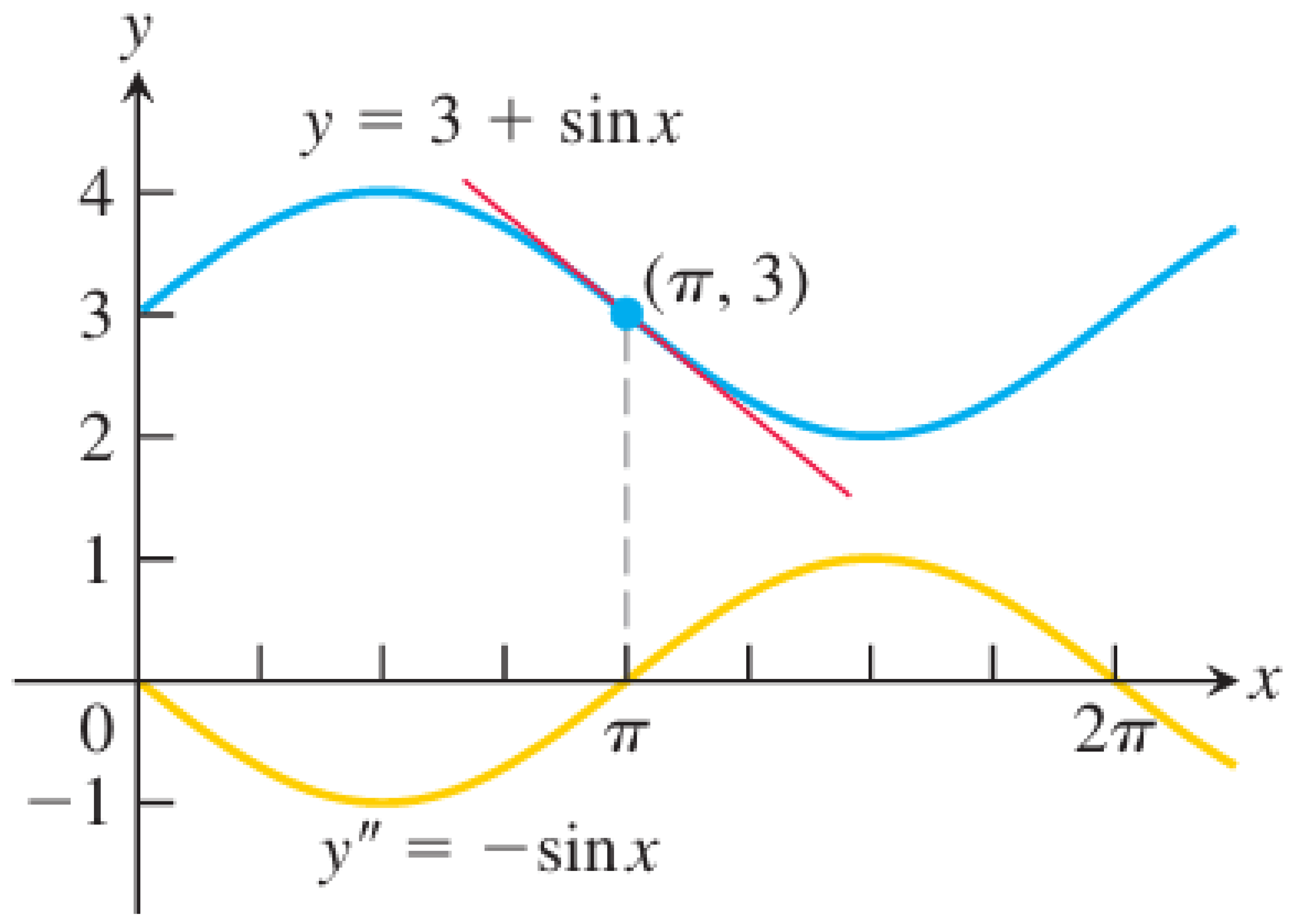


Definition 1.1.2 *A point $(c, f(c))$ where the graph of a function has a tangent line and where the concavity changes is a* ***point of inflection***

Example 1.1.2 *Determine the concavity of the function $y = 3 + \sin x$ on $[0, 2\pi]$.*

Sol

The first derivative of $y = 3 + \sin x$ is $y' = \cos x$, and the second derivative is $y'' = -\sin x \Rightarrow$ The graph of $y = 3 + \sin x$ is concave down on $(0, \pi)$, where $y'' = -\sin x$ is negative. And it is concave up on $(\pi, 2\pi)$ where $y'' = -\sin x$ is positive.



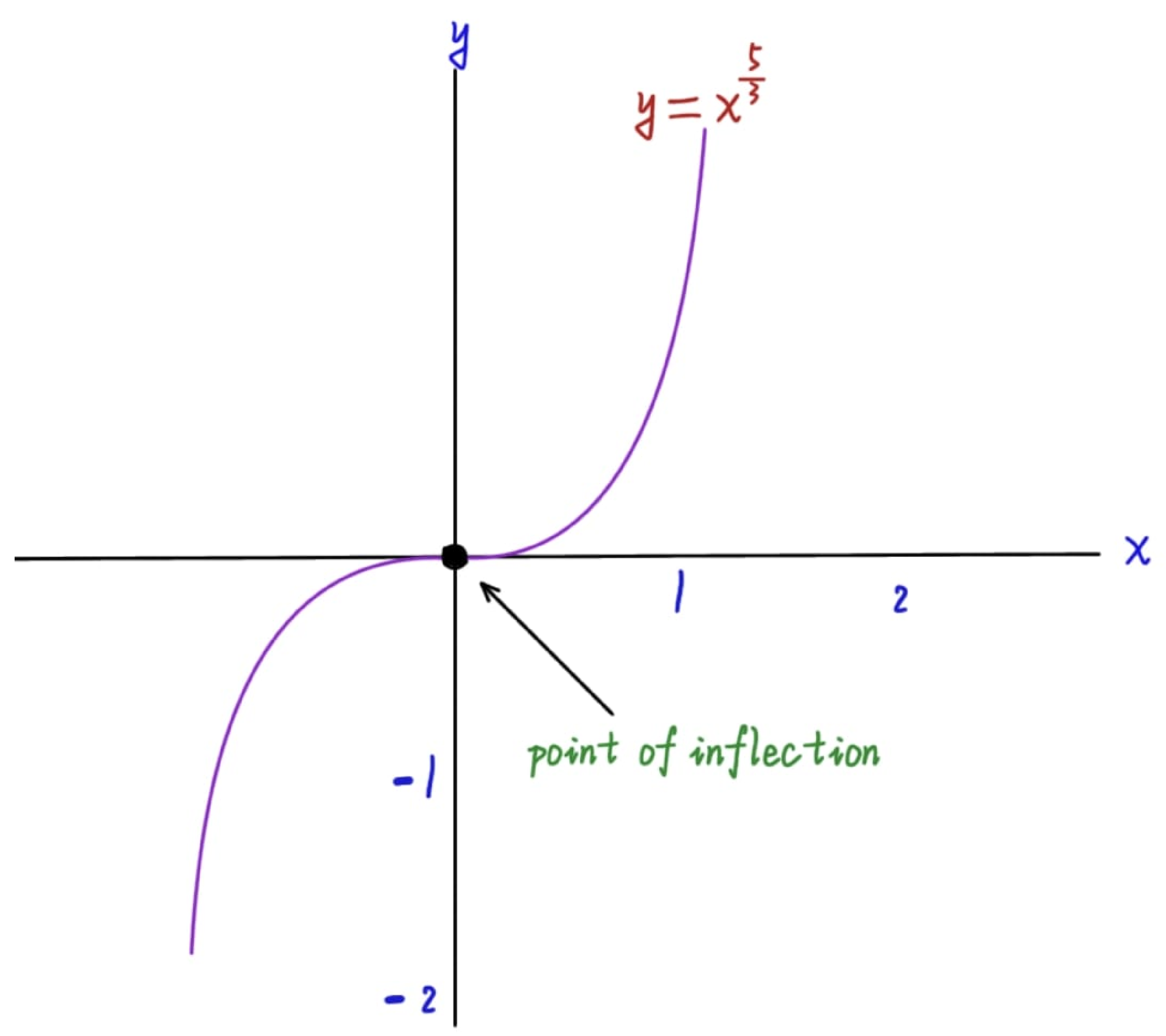
Example 1.1.3 *Determine the concavity and the inflection points of the function $f(x) = x^3 - 3x^2 + 2$*

Sol

$f' = 3x^2 - 6x$ and $f''(x) = 6x - 6$. To determine the concavity, we look at the sign of the second derivative $f''(x) = 6x - 6$. The sign is negative when $x < 1$, is 0 at $x = 1$ and is positive when $x > 1$. It follows that the graph of f is concave down on $(-\infty, 1)$, is concave up on $(1, \infty)$, and has an inflection point at $(1, 0)$ where the concavity changes.

Note: We did not need to know the shape of this graph ahead of time in order to determine its concavity.

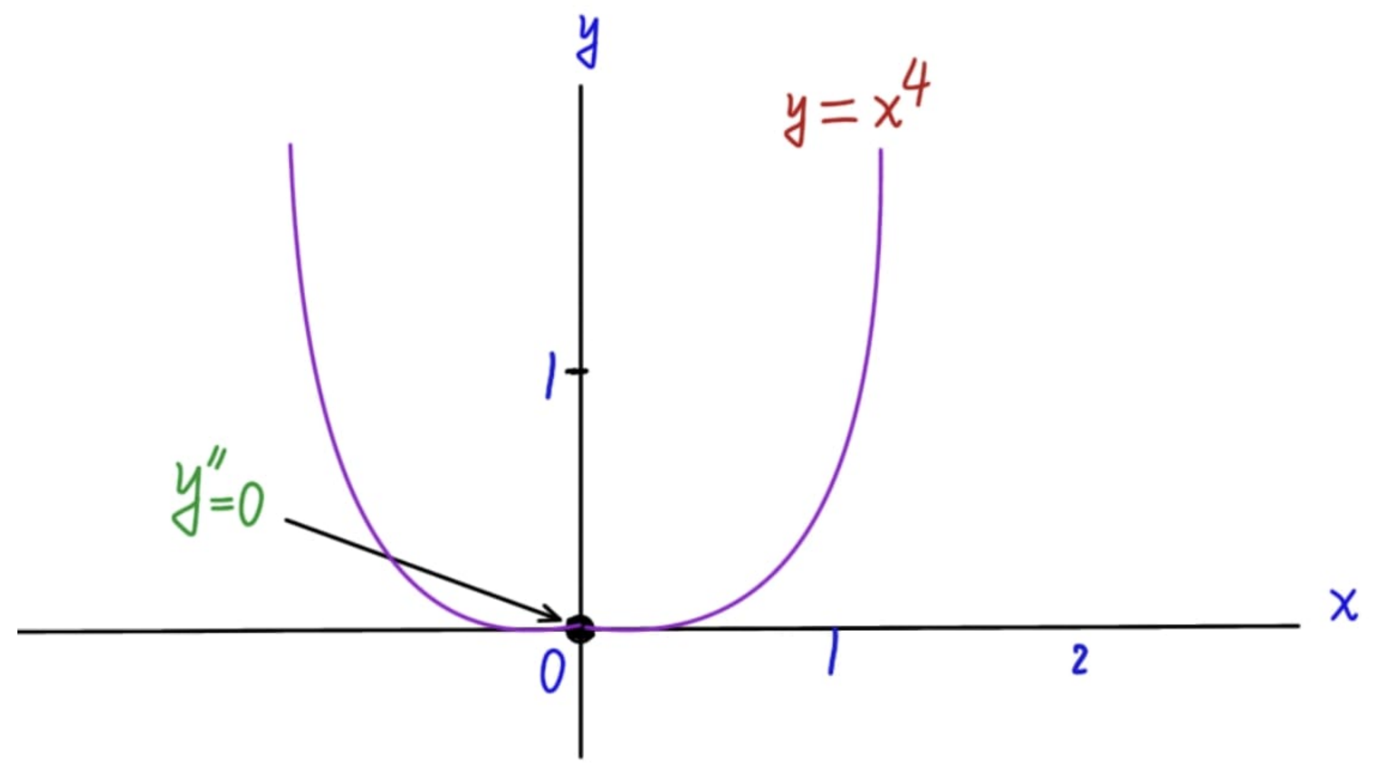
Example 1.1.4 *This example illustrates that a function can have a point of inflection where first derivative exists but the second derivative fails to exist. The graph of $f(x) = x^{\frac{5}{3}}$ has a*



horizontal tangent at the origin because $f^{'}(x) = \frac{5}{3}x^{\frac{2}{3}} = 0$ *when* $x = 0$. *However, the second derivative* $f^{''}(x) = \frac{d}{dx}\left[\frac{5}{3}x^{\frac{2}{3}}\right] = \frac{10}{9}x^{\frac{-1}{3}}$ *fails to exist at* $x = 0$.

Nevertheless, $f^{''}(x) < 0$ *for* $x < 0$ *and* $f^{''}(x) > 0$ *for* $x > 0$, *so the second derivative changes sign at* $x = 0$ *and there is a point of inflection at the origin.*

Example 1.1.5 *The curve* $y = x^4$ *has no inflection point at* $x = 0$. *Even though the second derivative* $y^{''} = 12x^2$ *is zero there, it does not change sign. The curve is concave up every where.*

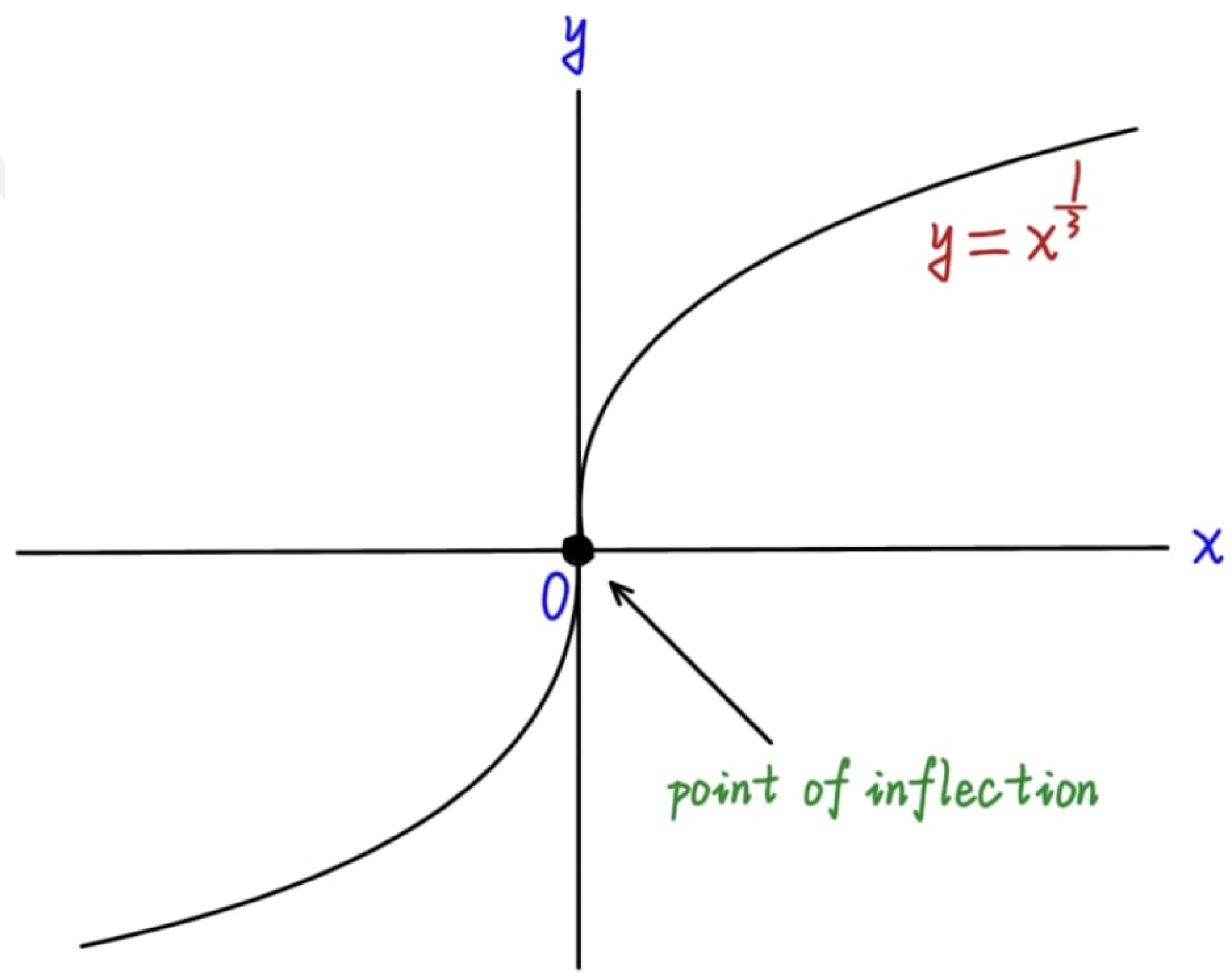


Example 1.1.6 *In this example, a point of inflection occurs at a vertical tangent to the curve where neither the first nor the second derivative exists.*

The graph of $y = x^{\frac{1}{3}}$ *has a point of inflection at the origin because the second derivative is positive for* $x < 0$ *and a negative for* $x > 0$.

$$y^{''} = \frac{d^2}{dx^2}\left[x^{\frac{1}{3}}\right] = \frac{d}{dx}\left[\frac{1}{3}x^{-\frac{2}{3}}\right] = -\frac{2}{9}x^{-\frac{5}{3}}.$$

However, both $y^{'}$ *and* $y^{''}$ *fails to exist at* $x = 0$, *and there is a vertical tangent there.*



Example 1.1.7 *A particle is moving along a horizontal coordinate line (positive to right) with position function*

$$s(t) = 2t^3 - 14t^2 + 22t - 5$$

Find the velocity and acceleration and describe the motion of the particle.

Sol

The velocity is

$$v(t) = s'(t) = 6t^2 - 28t + 22$$

and the acceleration is :

$$a(t) = v'(t) = s''(t) = 12t - 28 = 4(3t - 7)$$

when the function $s(t)$ is increasing, the particle is moving to the right. When $s(t)$ is decreasing, the particle is moving to the left. Note that v is zero at the critical points $t = 1$, $t = \frac{11}{3}$.

Intervals	$0 < t < 1$	$1 < t < \frac{11}{3}$	$\frac{11}{3} < t$
Sign of v	$+$	$-$	$+$
Behaviour of s	*increase*	*decrease*	*increase*
Particle motion	*right*	*left*	*right*

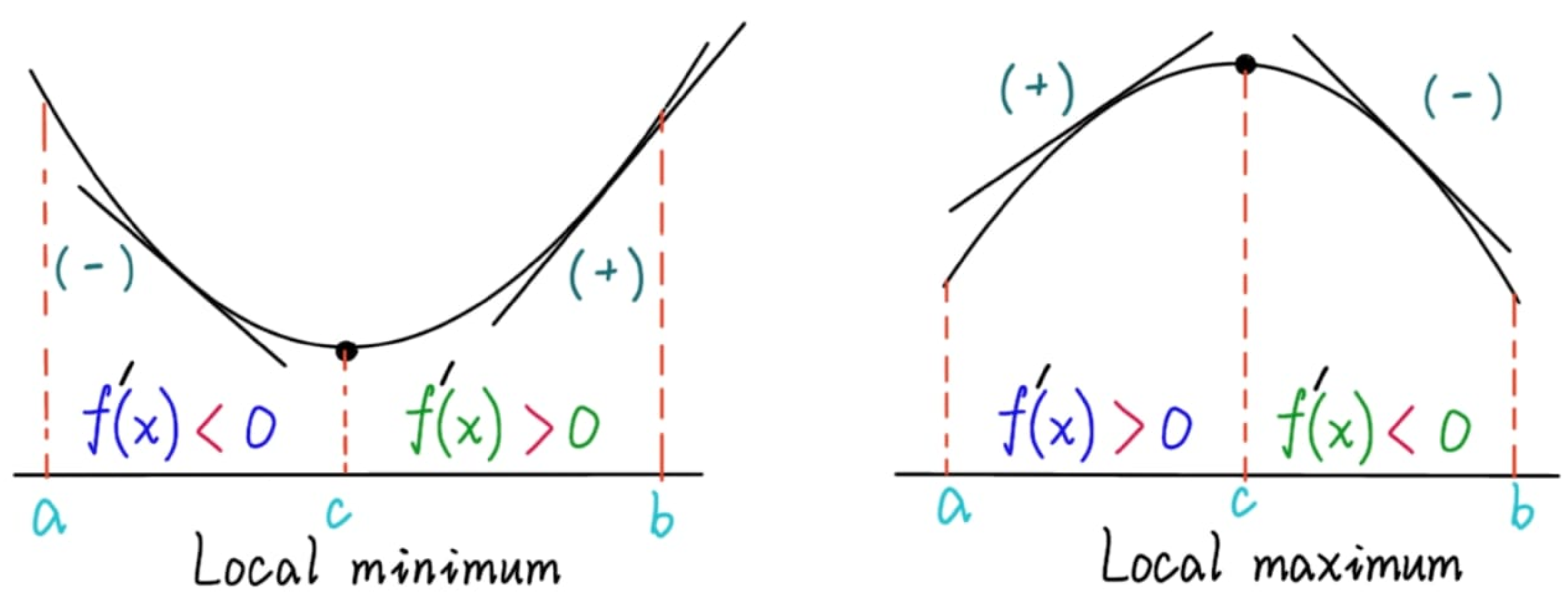
The particle is moving to the right in the time intervals $[0, 1)$ and $\left(\frac{11}{3}, \infty\right)$ and moving to the left in $\left(1, \frac{11}{3}\right)$. It is momentarily stationary (at rest) at $t = 1$ and $t = \frac{11}{3}$. The acceleration $a(t) = 4(3t - 7)$ is zero when $t = \frac{7}{3}$.

Intervals	$0 < t < \frac{7}{3}$	$\frac{7}{3} < 1$
Sign of $a(t)$	$-$	$+$
Graph of $s(t)$	*Concave down*	*Concave up*

The particle starts out moving to the right while slowing down and then reveres and begins moving to the left at $t = 1$ under the influence of the leftward acceleration over the time interval $\left[0, \frac{7}{3}\right)$ then $a(t)$ changes direction at $t = \frac{7}{3}$ but the particle continues moving leftward, while slowing down under the rightward acceleration. At $t = \frac{11}{3}$ the particle reverses direction again moving to the right in the same direction as the acceleration, so it is speeding up.

1.2 The Second Derivative Test:

It can be used to perform a simple test for relative maxima and minima. The test is based on the fact that "if the graph of a function f is concave upward on an open interval containing c, and $f'(c) = 0$, then $f(c)$ must be a relative minimum of f. Similarly, if the graph of a function f is concave downward on an open interval containing c, and $f'(c) = 0$, then $f(c)$ must be a relative maximum of f".



Theorem 1.2.1 ***Second Derivative Test:***

Let f be a function such that $f^{'}(c) = 0$ and its second derivative exists on an open interval containing c.

1. *If $f^{''}(c) > 0$, then $f(c)$ is relative minimum.*
2. *If $f^{''}(c) < 0$, then $f(c)$ is a relative maximum.*
3. *If $f^{''}(c) = 0$, then the test fails in such case, and we can use the first derivative test.*

Example 1.2.8 *Find the relative extrema for $f(x) = -3^5 + 5x^3$*

Sol

Begin by finding the critical points of f.

$$f^{'}(x) = -15x^4 + 15x^2 = 15x^2(1 - x^2) = 0 \Rightarrow \boxed{x = -1, 0, 1}$$

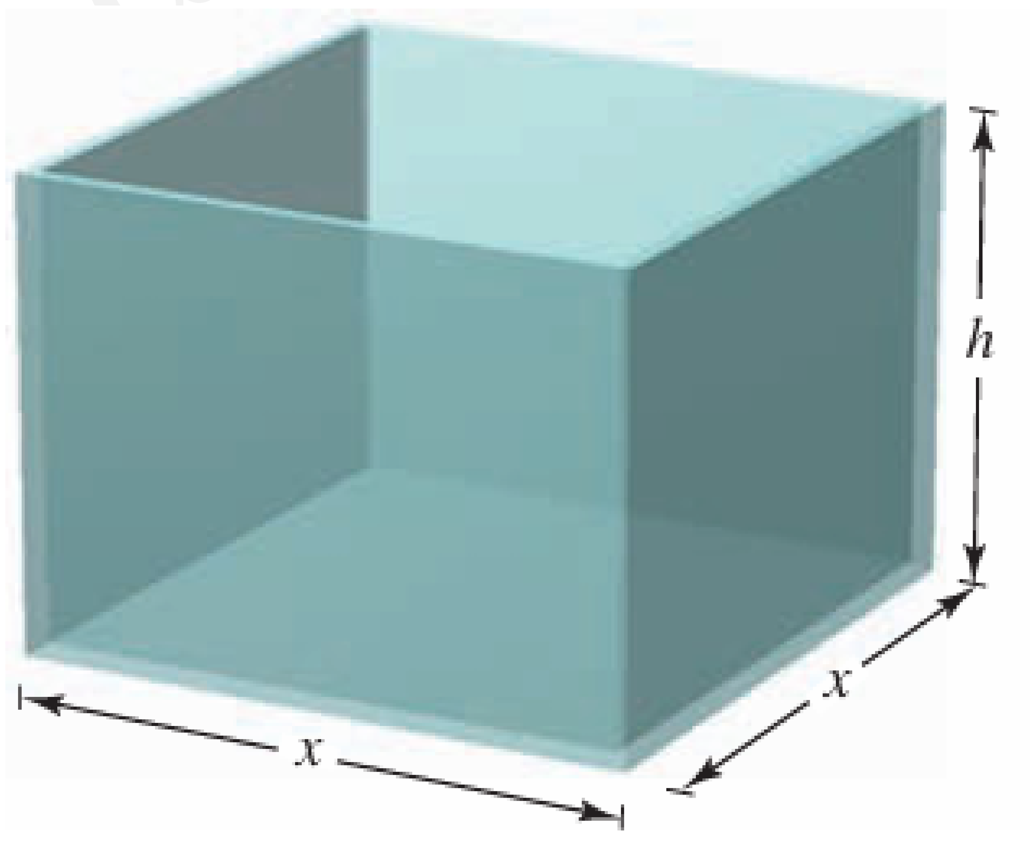
Now $f^{''}(x) = -60x^3 + 30x = 30(-2x^3 + x)$. We can apply the second derivative test as follows.

Point	*Sign of $f^{''}$*	*Conclusion*
$(-1,-2)$	$f^{''}(-1) = 30 > 0$	*relative minimum*
$(1,2)$	$f^{''}(1) = -3 < 0$	*relative maximum*
$(0,0)$	$f^{''}(0) = 0$	*Test fails*

1.3 Applied Min and Max Problems

Example 1.3.9 *A manufacturer wants to design an open box having a square base and a surface area of $108\ in^2$, what dimension will produce a box with maximum volume?*

Sol



Because the box has a square base, its volume is

$$V = x^2 h \qquad\qquad \textit{primary equation}$$

The surface of the box is $S =$ (area of the box)+(area of the four sides). Now:

$$S = x^2 + 4xh = 10$$

Because V is to be maximized, we want to express V as a function of one variable.

$$\Rightarrow h = \frac{108 - x^2}{4x}$$

Now $V = x^2h = x^2\left(\dfrac{108 - x^2}{4x}\right) = 27x - \dfrac{x^3}{4}$. *The domain of* f *(to be positive),* $V \geq 0$

$$27x - \frac{x^3}{4} \geq 0 \Rightarrow \frac{x^3}{4} \leq 27x \Rightarrow x^2 \leq 108 \Rightarrow 0 \leq x \leq \sqrt{108} \ (feasible\ domain)$$

To maximize V*, we can find the critical numbers of the volume function.*

$$\frac{dV}{dx} = 27 - \frac{3x^2}{4} = 0 \Rightarrow 3x^2 = 108 \Rightarrow x^2 = 36 \Rightarrow x = \pm 6.$$

Evaluating V *at* $(x = 6)$ *in the domain, and the endpoints of the domain produces* $V(0) = 0$ *and* $V(6) = 108$ *and* $V\left(\sqrt{108}\right)$.

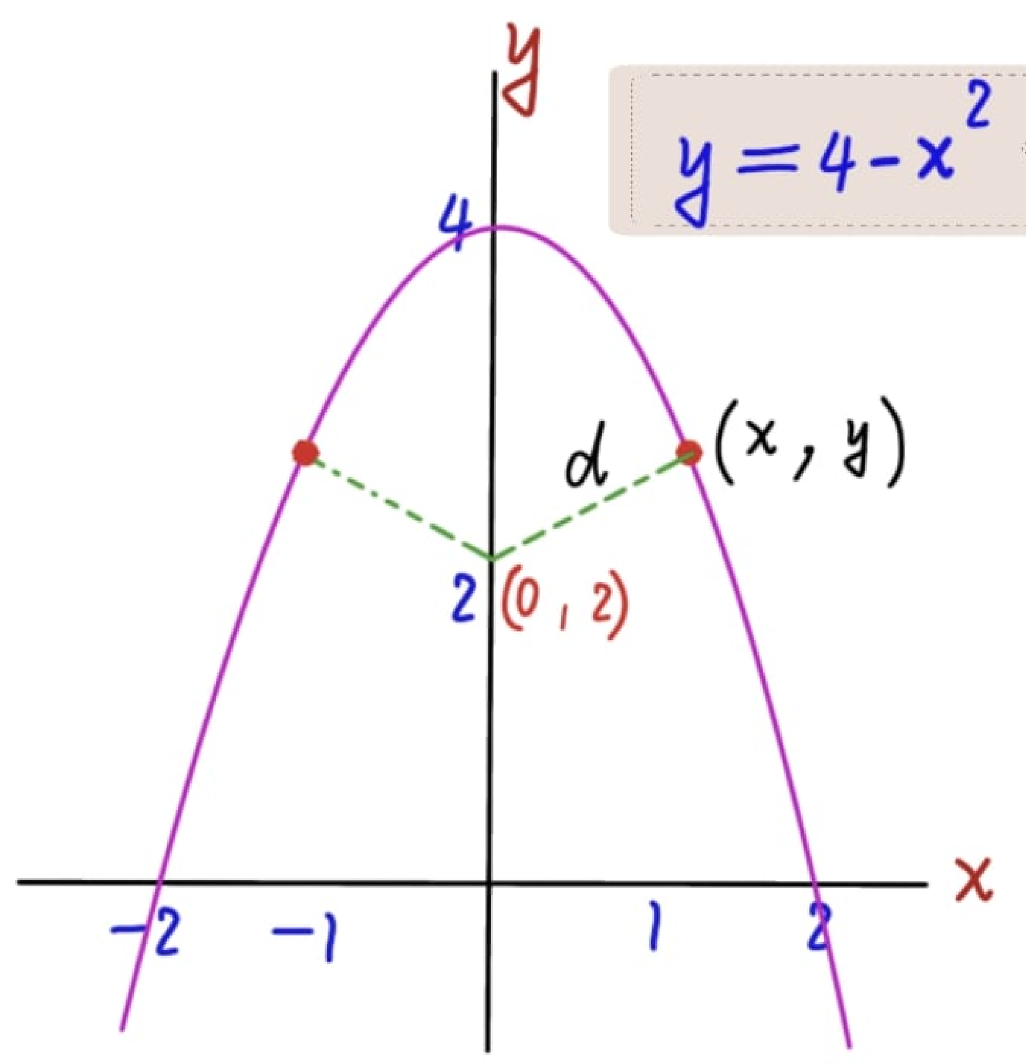
$\Rightarrow V$ *is maximum when* $x = 6$ *and the dimensions of the box are* $6 \times 6 \times 3$ *inches.*

1.3.1 Problem Solving Strategy for Applied Min and Max Problems

1. Assign symbols to all given quantities to be determined. When feasible, make a sketch.
2. Write a **primary equation** for the quantity that is to be maximized (or minimized).
3. Reduce the primary equation to one having a single independent variable. This may involve the use of the secondary equation relating the independent variables of the primary equation.
4. Determine the domain of the primary equation for which the stated problem make sense.
5. Use method of calculus to find the desired maximum or minimum value.

Example 1.3.10 *Which point on the graph of* $y = 4 - x^2$ *are closest to the point* $(0, 2)$*?*

Sol



The figure indicates that there are two points at a minimum distance from the point $(0, 2)$*. The distance between the point* $(0, 2)$ *and a point* (x, y) *on the graph of* $y = 4 - x^2$ *is given by*

$$d = \sqrt{(x - 0)^2 + (y - 2)^2} \qquad primary$$

Using the secondary equation $y = 4 - x^2$ *we can rewrite the primary equation as:*

$$d = \sqrt{x^2 + (4 - x^2 - 2)^2} = \sqrt{x^4 - 3x^2 + 4}$$

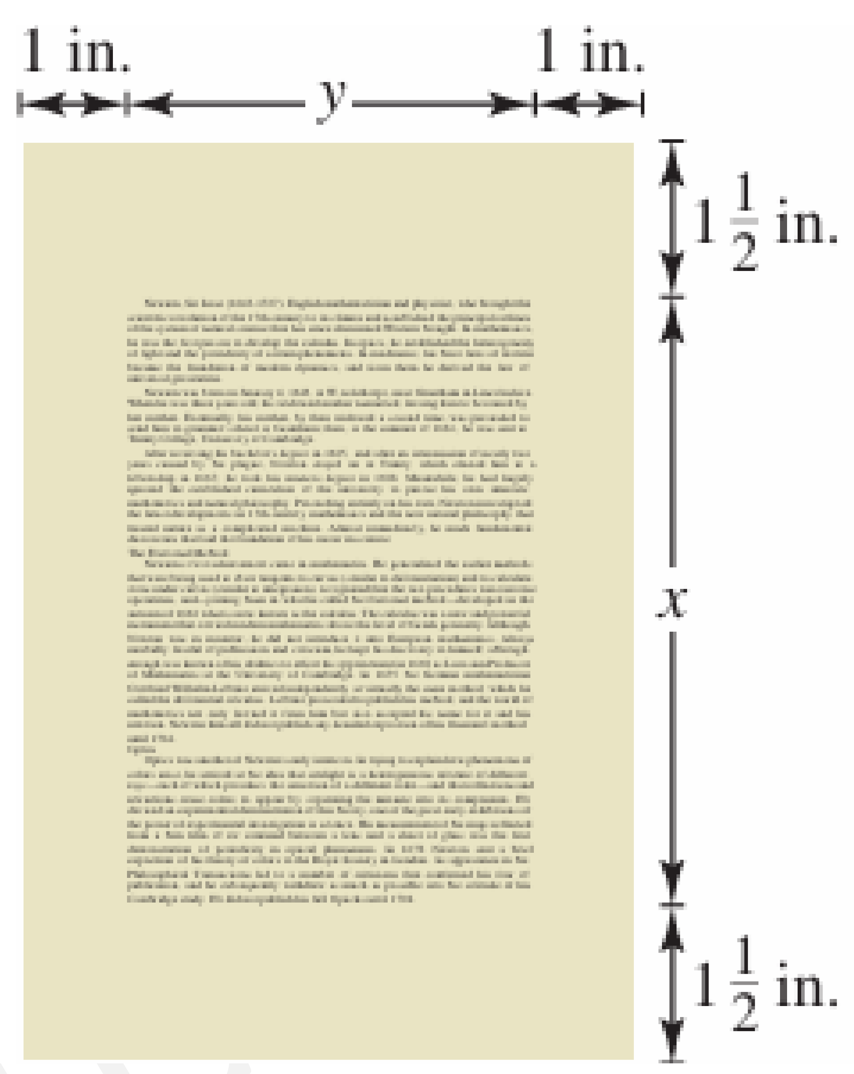
Here we need only to find the critical numbers of $f(x) = x^4 - 3x^2 + 4$*, and note that the domain of* f *is the entire real line.*

$$f^{'}(x) = 4x^3 - 6x = 2x(2x^2 - 3) = 0 \Rightarrow x = 0, \sqrt{\frac{3}{2}}, -\sqrt{\frac{3}{2}}$$

The first derivative test verifies that $x = 0$ *yields a relative maximum, where as both* $x = \sqrt{\frac{3}{2}}, -\sqrt{\frac{3}{2}}$ *yield a minimum distance. Hence the closest points are* $\left(\sqrt{\frac{3}{2}}, \frac{5}{2}\right)$ *and* $\left(-\sqrt{\frac{3}{2}}, \frac{5}{2}\right)$.

Example 1.3.11 *A rectangular page is to contain* $24\ in^2$ *of print. The margins at the top and bottom of the page are to be* $1\frac{1}{2}$ *inches, and the margins on the left and right are to be* 1 *inche. What should dimensions of the page be so that the least amount of paper is used?*

Sol



Let A *be the area to be minimized*

$$A = (x + 3)(y + 2) \qquad \textit{primary}$$

The printed area inside the margins is given by $24xy$*. Solving this equation for* y *produces* $y = \frac{24}{x}$*. Substitute this equation into the primary equation equation produces:*

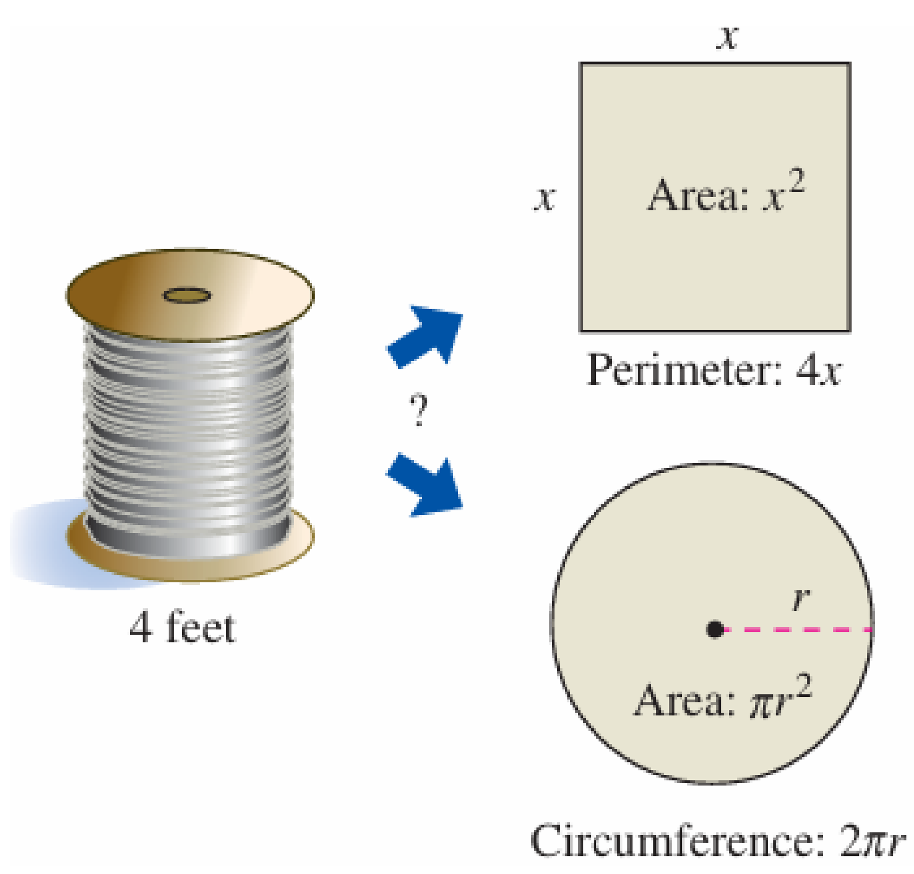
$$A = (x + 3)\left(\frac{24}{x} + 2\right) = 30 + 2x + \frac{72}{x}$$

Because x *must be positive, we are interested only in values of* A *for* $x > 0$*. To find the critical numbers, differentiate with respect to* x*.*

$$\frac{dA}{dx} = 2 - \frac{72}{x^2} = 0 \Rightarrow x^2 = 36 \Rightarrow x = 6$$

The first derivative test confirms that A *is a minimum when* $x = 6$*. Therefore* $y = \frac{24}{6} = 4$ *and the dimension of the page should be* $x + 3 = 9$ *inches by* $y + 2 = 6$ *inches.*

Example 1.3.12 *Four feet of wire is to be used to form a square and a circle. How much of the wire should be used for the square and how much should be used for the circle to enclose the maximum total area?*



Sol

$A =$ *area of square + area of circle*

$$\Rightarrow A = x^2 + \pi r^2 \qquad primary$$

Because the total length of wire is 4 feet, we obtain

$$\begin{aligned} A =& (perimeter\ of\ square) + (circumference\ of\ circle) \\ \Rightarrow 4 =& 4x + 2\pi r \end{aligned}$$

So $r = \dfrac{4-4x}{2\pi} = \dfrac{2-2x}{\pi} = \dfrac{2(1-x)}{\pi}$. *Then*

$$A = x^2 + \pi\left(\frac{2(1-x)}{\pi}\right)^2 = x^2 + \frac{4(1-x)^2}{\pi} = \frac{1}{\pi}[(\pi+4)x^2 - 8x + 4]$$

The feasible domain is $0 \le x \le 1$ *restricted by the square's perimeter. Because*

$$\frac{dA}{dx} = \frac{2(\pi+4)x - 8}{\pi}$$

The only critical number in $(0,1)$ *is* $x = \dfrac{4}{\pi+4} \approx 0.56$. *So, using* $A(0) \approx 1.273$, $A(0.56) \approx 0.56$ *and* $A(1) = 1$.

We conclude that the maximum area occurs when $x = 0 \Rightarrow$ *all the wire is used for the circle.*